

3D Modeling: Surface Area & Volume (Think First, Then Compute)

Mini-Review (use these wisely)

Rectangular prism (box): $V = \ell wh$, **Cone (right):** $V = \frac{1}{3}\pi r^2 h$; Lateral = πrs ; Total = $\pi r(r + s)$

Right square pyramid: $V = \frac{1}{3}Bh = \frac{1}{3}a^2 h$; Lateral = $\frac{1}{2}(4a)s = 2as$; Total = $a^2 + 2as$ **Sphere:** $V = \frac{4}{3}\pi r^3$, $SA = 4\pi r^2$

Cylinder: $V = \pi r^2 h$; Lateral = $2\pi rh$; Total = $2\pi r(h + r)$ **Hints:** Slant s for a right pyramid/cone is along a face. For right square pyramid, $s = \sqrt{h^2 + (a/2)^2}$. For cone, $s = \sqrt{h^2 + r^2}$. Units matter.

1. (**Volume first, then SA**) A rectangular prism has volume $V = 864 \text{ cm}^3$ with base $\ell \times w = 12 \text{ cm} \times 6 \text{ cm}$. Find h and then the total surface area.

2. (**Open-top box modeling**) A gift box is a rectangular prism with no top: $\ell = 18 \text{ cm}$, $w = 12 \text{ cm}$, $h = 10 \text{ cm}$. How much cardboard (area) is needed? Then how much ribbon covers a single loop around the middle (once around the sides, no overlap)?

3. (**Equal volume, different shapes**) A cylinder of radius $r = 4 \text{ cm}$ and height h has the *same volume* as a cone of radius 4 cm and height 18 cm. Find h . Which has greater surface area (lateral only)—justify without plugging giant numbers.

4. (**Find height from slant**) A right circular cone has radius $r = 6$ cm and slant height $s = 10$ cm. (a) Find the height. (b) Find its total surface area. (c) Find its volume.

5. (**Label wrap**) A paper label of height 8 cm wraps perfectly once around a straight cylinder with no gap/overlap. The label area is 264 cm^2 . Find the cylinder's radius and height.

6. (**From surface area to volume**) A sphere has surface area $SA = 324\pi \text{ cm}^2$. Find the radius, then its volume.

7. (**Square pyramid by slant**) A right square pyramid has base edge $a = 14$ cm and slant height $s = 13$ cm. (a) Find the total surface area. (b) Find the height h and volume.

8. (**Compare designs**) Two soup cans must hold the same volume $\pi r^2 h = 960\pi \text{ cm}^3$. Can

A has $r = 6$ cm; Can B has $r = 8$ cm. Find each height. Which design uses less lateral area $2\pi rh$?

9. (**Cone $\rightarrow$ cylinder transfer**) A full cone (radius 5 cm, height 12 cm) is poured into a cylinder of radius 5 cm. How high does the water rise?

10. (**Find a missing base edge**) A rectangular base has area $A = 144$ cm² and one side $w = 8$ cm. The box's height is $h = 15$ cm. Find total surface area.

11. (**Paint coverage**) You paint the *outside* of a closed cylindrical tank of radius $r = 3$ m and height $h = 5$ m. One gallon covers 28 m². How many gallons (to the next whole) are needed?

12. (**Hemisphere + cylinder**) A solid is a cylinder of radius 6 cm and height 10 cm *topped* with a hemisphere of radius 6 cm. (a) Find the total volume. (b) Find the exterior surface area (include the hemisphere and the curved side of the cylinder, and

the bottom circle; the circle where they meet is interior—do not count it).

13. (**Inscribed sphere**) A sphere fits perfectly inside a cube (touching each face). If the cube's volume is 512 cm^3 , find the sphere's surface area and volume.

14. (**Same base and height**) A right square pyramid and a right circular cone have *the same* base area B and height h . If the cone's volume is 120 cm^3 , what is the pyramid's volume? Explain the reasoning pattern without using B numerically.

15. (**Unit conversion & modeling**) A storage bin is a rectangular prism $\ell = 80 \text{ cm}$, $w = 50 \text{ cm}$, $h = 45 \text{ cm}$. (a) What is its capacity in liters? (b) A sticker says "Fill only to 85%". How many liters is that?

16. (**Packing spheres, simple model**) A box of interior $\ell \times w \times h = 30 \text{ cm} \times 20 \text{ cm} \times 10 \text{ cm}$ is filled with *one layer* of tennis balls (spheres) of diameter 10 cm , arranged in a simple

3×2 grid. (a) How many fit? (b) What fraction of the box's volume is actually tennis ball volume (use $r = 5$ cm)?

17. (**Cylinder vs. sphere, same r**) A cylinder and a sphere have the same radius r . The cylinder's height equals its radius. Which has greater volume? Justify with formulas (no numbers).

18. (**Find an unknown from equal surface areas**) A closed cylinder and a sphere have the *same* surface area. The cylinder's radius is r and its height is $2r$. Find the sphere's radius in terms of r .
